

Week 1: Data Collection, Preprocessing, and Initial Analysis

Day 1: Project Setup and Data Collection

Objectives:

- Set up the project repository and version control.
- Download and import project data.

Deliverables:

- Set up a GitHub repository for version control.
 - Collect data.
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Day 2: Data Preprocessing - Cleaning and Normalizing

Objectives:

- Clean the collected data by handling missing values, removing duplicates, and correcting any inconsistencies.
- Normalize and format the data for further analysis.

Deliverables:

- Cleaned datasets with missing values handled (e.g., imputation or removal) and outliers addressed.
 - Standardized time periods, product names, and customer IDs.
 - Initial report on the state of the data (quality checks, issues found).
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Day 3: Data Preprocessing - Feature Engineering

Objectives:

- Understand data structure and identify important features to be engineered (this process is iterable - meaning you can always come back, iterate and redo based on

business and model need).

- Finalize the formatting of the data for analysis and modeling.

Deliverables:

- Engineer features, ready for analysis.
 - Documentation of preprocessing steps. Properly document your preprocessing and feature engineering steps and your reason behind it.
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Day 4: Exploratory Data Analysis (EDA) - Initial Insights

Objectives:

- Conduct initial exploratory data analysis to uncover trends, seasonality, and correlations in the sales data.
- Visualize patterns in sales, promotions, and customer behavior.

Deliverables:

- EDA report with visualizations (e.g., line charts, histograms, box plots) for key trends.
 - Summary of findings, including seasonality trends, high-performing products, and customer behaviors.
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Day 5: Exploratory Data Analysis (EDA) - Deep Dive

Objectives:

- Perform a more in-depth analysis of the data, focusing on seasonality and correlations between sales, promotions, and external factors.

Deliverables:

- Deeper insights into seasonality, promotional impacts, and correlations with external factors like market conditions.
- Updated visualizations highlighting these deeper insights.

Week 2: Model Building and Evaluation

Day 6: ARIMA Model Implementation - Initial Setup

Objectives:

- Begin implementing the ARIMA model for instance, for forecasting based on historical sales data.
- Train the ARIMA model using the cleaned and formatted dataset.

Deliverables:

- ARIMA model implementation with initial hyperparameter settings.
- Initial training results, including model accuracy on historical data.

Day 7: ARIMA Model Tuning (Example model)

Objectives:

- Fine-tune the ARIMA model by adjusting hyperparameters like the number of lags and smoothing factors.
- Train the model on different subsets of the data for improved accuracy.

Deliverables:

- Refined ARIMA model with improved forecasting results.
- Documentation of hyperparameter tuning and its effects on model accuracy.

Day 8: Exponential Smoothing Model Implementation (Experimentation with a different model)

Objectives:

- Implement the Exponential Smoothing model for sales forecasting.

- Train the model and evaluate its initial performance.

Deliverables:

- Exponential Smoothing model implemented and trained on historical data.
 - Initial performance evaluation report comparing Exponential Smoothing with ARIMA.
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Day 9: Model Evaluation and Comparison

Objectives:

- Compare the performance of the ARIMA and Exponential Smoothing models on unseen data.
- Evaluate models using key metrics such as RMSE, MAE, and forecast accuracy.

Deliverables:

- Model evaluation report comparing ARIMA and Exponential Smoothing based on performance metrics.
 - Selection of the best-performing model so far for further refinement.
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Day 10: Random Forest Model Implementation

Objectives:

- Implement a Random Forest model to add another layer of comparison for forecasting accuracy.
- Train the Random Forest model using the available data.

Deliverables:

- Random Forest model implementation.
- Initial performance evaluation and comparison with ARIMA and Exponential Smoothing models.

Week 3: Final Model Selection, Deployment, and Visualization

Day 11: Model Comparison and Final Selection

Objectives:

- Finalize the selection of the best forecasting model (ARIMA, Exponential Smoothing, or Random Forest) based on performance metrics.
- Perform additional tuning and adjustments if needed.

Deliverables:

- Final model selection document based on comparison of all models (ARIMA, Exponential Smoothing, Random Forest).
- Refined version of the selected model with tuned hyperparameters.

Day 12: Docker Containerization for Deployment

Objectives:

- Prepare the model for deployment by packaging it into a Docker container.
- Ensure portability and consistency across different environments (e.g., development, staging, production).

Deliverables:

- Docker container for the selected model.
- End-to-end testing of the Dockerized model in a staging environment.

Day 13: Web Deployment - Streamlit Dashboard Setup

Objectives:

- Develop an interactive web application using **Streamlit** for visualizing the sales forecast.

- Integrate the forecasting model into the dashboard for user interaction.

Deliverables:

- Streamlit dashboard built, allowing users to input parameters and see sales forecasts.
 - Basic features like real-time forecasts, historical comparison, and trend visualization integrated into the dashboard.
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Day 14: Web Deployment - Final Testing and Refinements**Objectives:**

- Finalize the web deployment, testing for user interface issues, responsiveness, and dashboard functionality.
- Refine the model's user experience based on stakeholder feedback.

Deliverables:

- Streamlit dashboard deployed and tested.
 - User interface refinements and documentation on how to interact with the dashboard.
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Day 15: Project Review, Reporting, and Handoff**Objectives:**

- Conduct a final review of the entire project, ensuring all objectives have been met.
- Prepare a comprehensive report and presentation for stakeholders.

Deliverables:

- Final project report covering the methodology, model performance, deployment process, and business impact.
- Presentation slides summarizing key findings and recommendations for future improvements.

- Handoff document detailing the deployment process and next steps for monitoring and maintaining the forecasting system.